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Microbial driver of 2006–2023 CH_4 growth indicated by trends in atmospheric δD–CH_4 and $\delta^{13}C$–CH_4

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Methane (CH_4) is the second most important greenhouse gas and has been rising following a brief period of stabilization from 1999 to 2006. Determining the cause of this rise is critical for reducing emissions and predicting future climate sensitivity. The carbon and hydrogen stable isotopic composition of atmospheric CH_4 is controlled by variability in isotopically distinguishable emission categories and fractionating sink processes. While most studies using atmospheric $\delta^{13}C$–CH_4 data suggest a dominantly microbial source for recent CH_4 growth, this understanding is not uniform, and uncertainties remain [S. Schwietzke *et al.*, *Nature* 538, 88–91 (2016), S. Basu *et al.*, *Atmos. Chem. Phys.* 22, 15351–15377 (2022), J. Thanwerdas, M. Saunois, A. Berchet, I. Pison, P. Bousquet, *Atmos. Chem. Phys.* 24, 2129–2167 (2024)]. Here, we present a harmonized global measurement record of atmospheric δD–CH_4 and estimate emissions from 1999 to 2022 with global isotope mass balance calculations using both carbon and hydrogen isotopic ratios. We conduct thorough uncertainty analyses to separate absolute magnitude and emission trend uncertainties and find with high confidence that trends in $\delta^{13}C$–CH_4 and δD–CH_4 observations are both consistent with an entirely microbial emission driver of the post-2006 CH_4 rise, while fossil fuel emissions have remained relatively stable.

methane | isotope geochemistry | emissions | greenhouse gases | climate change

Significance

Methane is a potent greenhouse gas, and its accelerating rise since 2006 makes international climate goals harder to meet. However, there is debate about the specific sources driving this recent increase. By combining global records of methane's hydrogen and carbon isotopic compositions, this study provides strong evidence that microbial sources, such as those from wetlands, agriculture, and waste, are primarily responsible for the post-2006 methane rise, while fossil fuel emissions have remained relatively stable despite increased global production. Our findings highlight the importance of microbial methane sources and their potential sensitivities to changes in climate.

Methane (CH_4) is the second most important anthropogenic greenhouse gas contributing to contemporary climate change (1), yet substantial uncertainties remain in its global budget. In particular, the driver of recent atmospheric CH_4 growth following a brief stabilization from 1999 to 2006, determined from NOAA's Marine Boundary Layer (MBL) measurements, has been debated (e.g., refs. 2–4). Quantifying how the sources and sinks of CH_4 are changing over time is critical for predicting its future trajectory and informing mitigation strategies. The global CH_4 budget has been studied extensively through bottom–up and top–down approaches (e.g., refs. 4–7). Bottom–up estimates use inventories and process-based models to quantify emissions from natural and anthropogenic source sectors (8, 9), while top–down approaches use atmospheric observations to infer broader emission categories through inverse modeling and mass balance methods (5, 6, 10–15).

Despite advances in both methods, discrepancies in the role of microbial and fossil fuel emissions in recent CH_4 growth remain (3, 4, 14, 16, 17). Most isotope-enabled top–down modeling estimates agree that increasing microbial emissions are responsible for the large majority of the increase (5, 10, 15, 18). These studies strongly rely on constraints from the observed atmospheric trend of 13C:12C (expressed as $\delta^{13}C$–CH_4), which is sensitive to potential trends in the atmospheric oxidation of CH_4 by OH (2, 3) and tropospheric chlorine (Cl) (19), trends in source signatures (20), and variability in ^{13}C-enriched pyrogenic emissions (21).

Isotopic ratios of D:H in atmospheric CH_4 (expressed as δD–CH_4 or $\delta^{2}H$–CH_4) can provide additional, independent top–down constraints on the global CH_4 budget. δD–CH_4 and $\delta^{13}C$–CH_4 respond differently to CH_4 sources and sinks, with δD–CH_4 less affected by changes in the Cl sink and pyrogenic emissions but more sensitive to variations in wetland emissions and sink fractionation (22–26). Despite its potential, δD–CH_4 has not been fully integrated into global mass balance studies due to limited observations, and its role in understanding the CH_4 budget remains underexplored.

Here, we investigate the trend of atmospheric δD–CH_4 between 2005 and 2023, made possible by harmonizing a suite of datasets from different measurement programs. Using records of both global mean δD–CH_4 and $\delta^{13}C$–CH_4, we calculate two independent estimates of microbial and fossil fuel CH_4 emissions using a global mass balance framework. By employing a Monte Carlo (MC) uncertainty analysis that considers both

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correlated and uncorrelated uncertainties (Methods), we explicitly separate uncertainty in annual total fossil fuel and microbial emissions from the uncertainty in their long-term trends between 1999 and 2023. We find that the 140 ppb rise in atmospheric CH_4 and trends in both δD–CH_4 and δ^{13}C–CH_4 that occurred over this period are best explained by a microbial source, while fossil fuel emissions remained relatively stable. The strong agreement between δD–CH_4 and δ^{13}C–CH_4 mass balance calculations lends confidence to this conclusion, demonstrating the value of dual-isotopic constraints in refining the global CH_4 budget.

Results

Global δD–CH_4 Dataset. We use our harmonized global dataset of atmospheric δD–CH_4 (*SI Appendix*, Fig. S1) to calculate the global annual mean (Methods), which decreased by 6.0 ± 0.8‰ between 2005 and 2023, following a similar decreasing trend in δ^{13}C–CH_4 of 0.7‰ ± 0.02‰ across the same interval (Fig. 1; "±" denotes 1σ uncertainty hereafter). On top of these long-term trends, global mean δD–CH_4 and δ^{13}C–CH_4 exhibit clear seasonal cycling and interhemispheric differences. On average, Northern Hemisphere (NH) δD–CH_4 is ~12‰ lower than (Southern Hemisphere) SH δD–CH_4 (*SI Appendix*, Fig. S2), reflecting dominant NH emissions and the relatively older age of SH CH_4, which has undergone more oxidation that preferentially removes the lighter isotopologues. This interhemispheric difference is lowest during NH summer (~8‰) and highest during NH winter (~15‰) due to seasonal variability in the dominant OH sink (20, 22, 27).

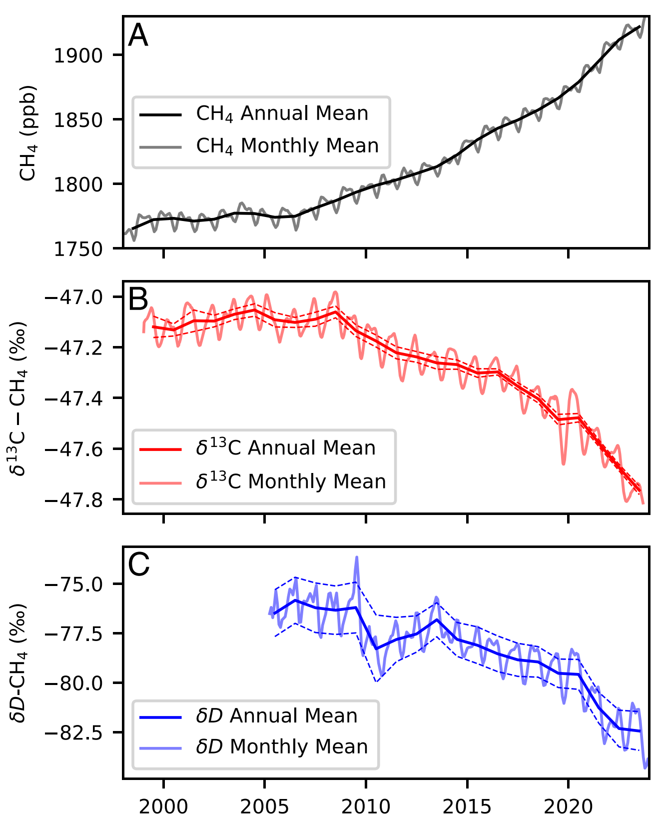

Fig. 1. Global mean CH_4, δ^{13}C–CH_4, and δD–CH_4. (*A*) Global mean annual (solid black line) and monthly (solid gray line) atmospheric CH_4 mixing ratio determined from NOAA's MBL measurements (28). (*B*) Global mean annual (solid dark red line) and monthly (solid light red line) atmospheric δ^{13}C–CH_4 plotted with annual two-sigma uncertainty windows (dashed red lines) (18). (*C*) Global mean annual (solid dark blue line) and monthly (solid light blue line) atmospheric δD–CH_4 plotted with annual two-sigma uncertainty windows (dashed blue lines). See Methods for details on global mean calculations.

While we focus here on global trends, the latitudinal and seasonal signals will be investigated in future studies and demonstrate the sensitivity of δD–CH_4 to variations in both sources and sinks.

Global δD–CH_4 and δ^{13}C–CH_4 Mass Balance. We use our annual mean atmospheric δD–CH_4 record and the well-established atmospheric CH_4 and δ^{13}C–CH_4 records (18, 28) in a global annual mass balance framework to solve for the magnitude and flux-weighted δD–CH_4 and δ^{13}C–CH_4 signatures of global annual emissions, given sink strength and fractionation factors (Table 1 and *SI Appendix*, Table S3; Methods). We then partition total CH_4 emissions into microbial and fossil fuel components using updated developed estimates of global annual mean source signatures (Methods) and annual pyrogenic emissions from the Global Fire Emissions Database (GFEDv4.1 s) (29). Global microbial and fossil emissions were smoothed using a 5-y smoothing window due to uncertainties in box model constraints on estimates of interannual emission variability (30). Global mean fossil, microbial, and pyrogenic source signatures and their trends over time are calculated with an updated compilation of source signature measurements (*SI Appendix*, Table S2) considering important known spatial (*SI Appendix*, Fig. S3) and temporal (Table 1 and *SI Appendix*, Figs. S4–S8) variability (Methods).

To quantify flux uncertainty, we use a 1,000-iteration MC framework in which each simulation uses new time series of global mean source signatures and atmospheric δ^{13}C–CH_4 or δD–CH_4 based on the mean and trend uncertainties (Table 1). Importantly, the mean and trend of each input are varied independently with respect to their uncertainties, allowing us to distinguish between the uncertainty in emissions for any given year and the uncertainty in emission trends over time. This distinction is critical because temporal trends (or lack thereof) in source signatures are better constrained than their absolute values (*SI Appendix*, Fig. S8), leading to substantially lower uncertainties in trend estimates than in flux magnitude (Fig. 2 and *SI Appendix*, Fig. S9). This enables more precise attribution of post-2006 atmospheric CH_4 growth to specific source changes. Through this MC error propagation and additional sensitivity tests (see below), we assess all major sources of uncertainty in our isotopic mass balance approaches and their influence on interpretations of the observed atmospheric δ^{13}C–CH_4 and δD–CH_4 trends.

We find that our δ^{13}C–CH_4 and δD–CH_4 mass balance calculations agree well in both their magnitudes and temporal trends (Fig. 2). Both indicate that the 140 ppb rise in atmospheric CH_4 between 2006 and 2023 was overwhelmingly driven by microbial emissions (δ^{13}C: 73 ± 5 Tg yr^{-1}; δD: 73 ± 12 Tg yr^{-1}). Only 16% (δ^{13}C) and 25% (δD) of fossil fuel emission ensemble members increase during this time (Figs. 2 *C* and *D* and 3 *A* and *B* and *SI Appendix*, Fig. S9). In fact, both datasets suggest a fossil emissions decrease between 2013 and 2022 (δ^{13}C: −14 ± 4 Tg yr^{-1}; δD: −15 ± 11 Tg yr^{-1}) following a slight increase between the 2005 to 2007 average and 2013 (δ^{13}C: +12 ± 2 Tg yr^{-1}; δD: +11 ± 7 Tg yr^{-1}), with long-term mean values of 160 ± 29 Tg yr^{-1} and 133 ± 33 Tg yr^{-1} for δ^{13}C and δD, respectively. Uncertainties in estimated emission trends are higher when using δD–CH_4 as a constraint (Fig. 2) due to greater uncertainties in the laboratory implementation of the V-SMOW calibration scale (24, 33), which should reduce once ongoing δD–CH_4 interlaboratory comparison efforts are completed (*SI Appendix*, Fig. S1; Methods). Despite including conservative uncertainty estimates in our mass balance, all simulations for both isotopes result in a far greater increase in microbial emissions than in fossil fuels. This agreement, along with consistent microbial and fossil fuel emission trends for the separate mass balance MC simulation ensembles (Fig. 3 *A* and *B*),

Table 1. Summary of uncertainties used in the isotope mass balance MC error propagation

Isotope	Model components	Mean	Mean 1σ Unc.	Trend	Trend 1σ Unc.
$\delta^{13}C$	Atmosphere	−47.3‰	±0.01‰	−0.66‰	±0.02‰
	Fossil Source	−44.0‰	±0.2‰	+0.60‰	±0.14‰
	Microbial Source	−62.0‰	±1.3‰	−0.37‰	±0.09‰
	Pyrogenic Source	−23.1‰	±2.5‰	0‰	0‰
	Net sink KIE	1.0082	0	0	0
δD	Atmosphere	−78.4‰	±0.5‰	−6.0‰	±0.77‰
	Fossil Source	−193.4‰	±0.6‰	+1.3‰	±0.7‰
	Microbial Source	−304.7‰	±8.2‰	+1.8‰	±0.5‰
	Pyrogenic Source	−228.0‰	±7.6‰	0‰	0‰
	Net sink KIE	1.281	0	0	0

Trends and means are presented as the global mean values and the change in those values between 1999 and 2023, respectively. For atmospheric δD, the range is 2005 to 2023. 0‰ indicates that either no trend or uncertainty is imposed in the baseline MC simulation. See Methods for calculation details. Sink-weighted net KIEs (*SI Appendix*, Table S3) are also shown. Global mean atmospheric isotopic composition and source signatures, temporal trends, and one-sigma uncertainties in the mean values and temporal trends for both isotopes.

provides strong evidence that rising microbial emissions have been the primary driver of post-2006 CH_4 growth.

Sensitivity Tests. A key advantage of using both $\delta^{13}C$–CH_4 and δD–CH_4 to constrain the global CH_4 budget is their differing sensitivities to key processes that influence atmospheric CH_4 isotopic composition. Interpretations of $\delta^{13}C$–CH_4 may be hindered by uncertainties and trends in the highly fractionating but poorly quantified Cl sink (*SI Appendix*, Table S3) (36, 37) and by potential trends in global pyrogenic CH_4 emissions, which are relatively more enriched in ^{13}C than D (Table 1) (21, 38). An unaccounted-for decline in either of these would cause some of the observed decrease in atmospheric $\delta^{13}C$–CH_4 to be misattributed to an increase in microbial emissions at the expense of fossil emissions. We explored these scenarios by solving our mass balance separately, assuming large reductions in the Cl sink and pyrogenic emissions from 2006 to 2022 (Methods). While the proposed trends are highly improbable, they are the conditions required for the $\delta^{13}C$ mass balance to attribute at least half of the post-2006 CH_4 rise to increased fossil fuel emissions. We find that atmospheric $\delta^{13}C$–CH_4 responds much more strongly to these changes than δD–CH_4.

Reducing the Cl sink over our study period from 3.5 to 1.0% of the total sink shifts the $\delta^{13}C$–CH_4 mass balance toward a fossil fuel emission-leaning contribution to the post-2006 CH_4 increase (+37 Tg yr^{-1} fossil, +31 Tg yr^{-1} microbial; Fig. 3 *C* and *D*). In contrast, the CH_4 increase remains microbially driven when using δD–CH_4 (+8 Tg yr^{-1} fossil, +57 Tg yr^{-1} microbial) (*SI Appendix*, Fig. S10). This statistically significant divergence is demonstrated by the minimal overlap in the suite of MC simulations of both isotopes (Fig. 3 *C* and *D*). In other words, if the negative trend in atmospheric $\delta^{13}C$–CH_4 were due to a reduced Cl sink, the expected decrease in atmospheric δD–CH_4 would be much

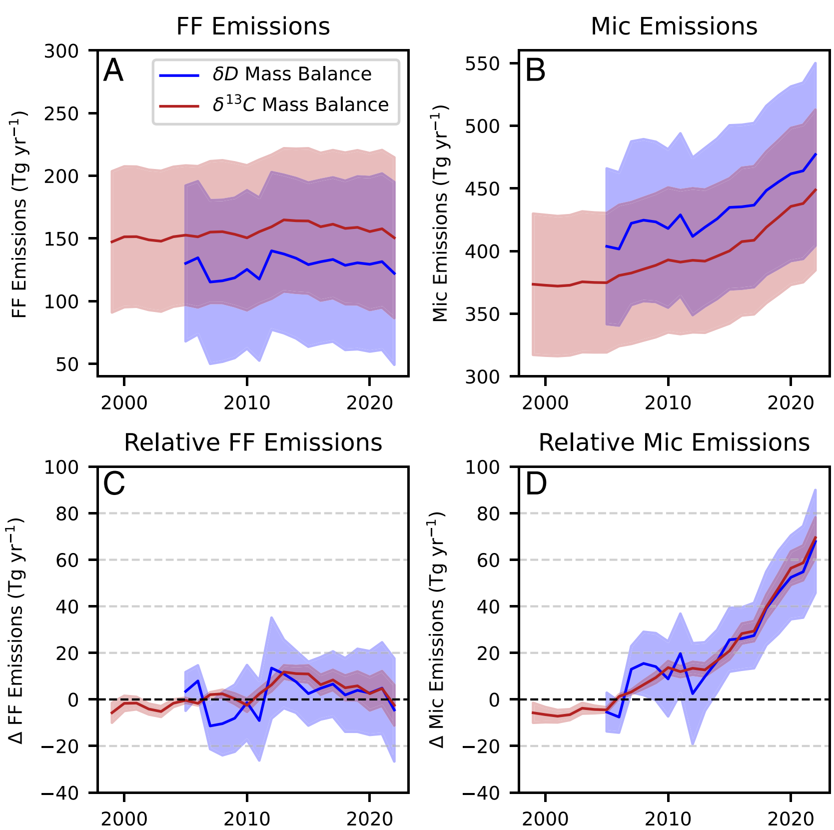

Fig. 2. Mass balance results. Smoothed global mass balance results from $\delta^{13}C$–CH_4 (red) and δD–CH_4 (blue). (*A*) Calculated total global fossil fuel (FF) CH_4 emissions and uncertainty [total fossil – 5 Tg yr^{-1} geologic emissions; (31, 32)] and (*B*) calculated total global microbial CH_4 emissions and uncertainty. Relative emissions and uncertainties are shown for fossil fuel (*C*) and microbial (*D*). Upper and lower bound two-sigma uncertainties are indicated by the shaded area.

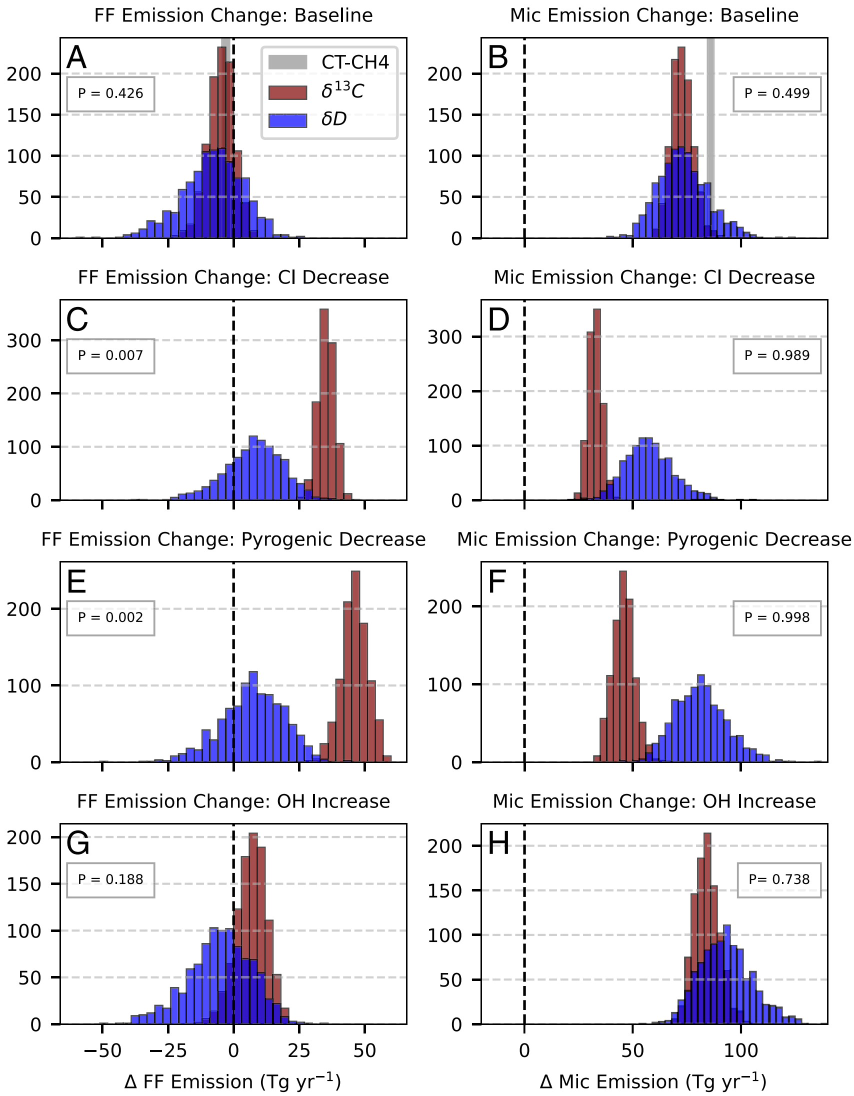

Fig. 3. Histogram of mass balance results for different sensitivity test scenarios. Histogram showing the change in fossil fuel (*Left*) and microbial (*Right*) emissions from the 2005 to 2007 average to the 2020 to 2022 average for all Monte Carlo runs of the δ^{13}C–CH_4 (red) and δD–CH_4 (blue) mass balances. No change is displayed as a black dashed line. (*A* and *B*) Best guess scenario presented in Fig. 2 and the CT-CH_4 posterior (gray bar). (*C* and *D*) Scenario where Cl sink decreases from 3.5 to 1.1% of the total sink. (*E* and *F*) Scenario where pyrogenic CH_4 emissions decrease to 2 Tg yr^{-1}. (*G* and *H*) Scenario where OH increases in line with recent studies (34, 35). *P* values represent the percent of MC iterations in which δD-derived emissions exceed δ^{13}C-derived emissions. $P < 0.05$ and $P > 0.95$ indicate that the change in δD-derived emissions is statistically smaller or larger than the change in δ^{13}C-derived emissions, respectively. This tests for statistically significant divergence in the two different trend results.

smaller, contradicting observations. Our isotopic results agree best if the Cl sink is held constant, with no statistically significant difference between the fossil and microbial trends derived from δ^{13}C–CH_4 and δD–CH_4 (Fig. 3 *A* and *B*). If the Cl sink has doubled in recent years, as suggested by (19), our isotopic results also diverge. This scenario would necessitate an additional 65 Tg yr^{-1} (δ^{13}C–CH_4) or 36 Tg yr^{-1} (δD–CH_4) increase in microbial CH_4 emissions over our study period, and a corresponding decline in fossil fuel emissions. Note that this effect would be muted if the Cl sink makes up a smaller proportion of the total sink (37).

An extreme and unrealistic reduction in pyrogenic emissions to 2 Tg yr^{-1} is required for trends in atmospheric δ^{13}C–CH_4 and post-2006 CH_4 growth to be mainly driven by increased fossil fuel emissions, as shown by (18). However, the δD–CH_4 mass balance results still show increased microbial emissions strongly driving CH_4 growth (+9 Tg yr^{-1} fossil, +82 Tg yr^{-1} microbial) and are thus inconsistent with the δ^{13}C–CH_4 results in this scenario (Fig. 3 *E* and *F*). The complete cessation of pyrogenic emissions is unrealistic, whereas recent studies have suggested 10 to 30% reductions over the last 20 y (21, 38). For a 30% reduction, fossil fuel emissions still decrease by 2 Tg yr^{-1} in the δD–CH_4 mass balance and only contribute to 13% of the post-2006 CH_4 rise in the δ^{13}C–CH_4 mass balance. Again, the mass balance results converge best when pyrogenic emissions are prescribed using GFED estimates (Fig. 3 *A* and *B*), providing indirect evidence that potential reductions in these emissions have not notably contributed to the observed decreases in atmospheric δ^{13}C–CH_4 or δD–CH_4 through 2023.

Although we hold CH_4 lifetime and sink partitioning constant in our mass balance calculations, there is some evidence that the dominant OH sink is increasing by ~0.3%/yr (34, 35). Incorporating this change into our mass balance (Methods) increases the magnitude of total emissions but minimally impacts the partitioning between microbial and fossil fuel emissions in both isotopic approaches (Fig. 3 *G* and *H*), despite variation in the difference between OH and net sink fractionation for δ^{13}C–CH_4 and δD–CH_4 (*SI Appendix*, Table S3). Specifically, if the OH sink were increasing in line with recent literature estimates (34), our mass balance would underestimate the increase in microbial emissions by at most 20 Tg yr^{-1}, with only minor changes to fossil fuel emissions.

Uncertainties in isotopic source signature trends have been cited as a disadvantage of isotopic constraints on the CH_4 budget (4, 11); however, this only appears to be an issue when significantly large trends are imposed. A recent dual-isotopic inversion study (11) explained CH_4 growth through 2016 with equal changes in fossil fuel and microbial emissions, allowing source signatures to vary annually within wide uncertainty bounds to fit atmospheric observations, leading to large long-term trends and interannual variability. Specifically, the inversion attributes much of the observed decline in atmospheric δ^{13}C–CH_4 to substantial decreases in the $\delta^{13}C–CH_4$ of both agriculture and waste (AGW) and fossil fuel emissions, resulting in a rising fossil fuel source. In contrast, we estimate global mean fossil fuel $\delta^{13}C–CH_4$ trends and uncertainties by flux-weighting spatial source signature maps with gridded emission fields (*SI Appendix*, Figs. S3 and S5; Methods), and find a statistically significant positive trend (*SI Appendix*, Fig. S12*B*), as expected due to the increasing fraction of oil and gas extraction from shale formations that are more enriched in ^{13}C (3, 39). For AGW, we identify a negative $\delta^{13}C–CH_4$ trend due to a combination of the Suess effect (20) and the changing isotopic composition of ruminant diets (40), though the magnitude is much smaller than the trend inferred by ref. 11 (*SI Appendix*, Fig. S12*A*; Methods). Unlike the inversion in ref. 11, which establishes source signature trends primarily from atmospheric $\delta^{13}C–CH_4$ variations, our estimates are empirically constrained by known geographic variability in source signatures and emissions (Methods). Further, our Monte Carlo framework accounts for uncertainty in these calculated temporal source signature trends (Methods), and the influence of these trends on modeled net emission changes is limited to ±10 Tg yr^{-1} (Methods; *SI Appendix*, Fig. S11).

Discussion

Existing top–down modeling studies that find a microbial driver of post-2006 CH_4 growth cite trends in atmospheric $\delta^{13}C–CH_4$ as the primary evidence (3, 5, 10, 16, 18, 41, 42). Our results show that trends in atmospheric $\delta D–CH_4$ are also consistent with a microbial driver, providing strong support to interpretations of $\delta^{13}C–CH_4$. The Global Carbon Project (GCP) does not currently incorporate atmospheric $\delta^{13}C–CH_4$ into its budget estimates, citing uncertainties in source signatures and sink fractionation (4). This concern is most relevant to the absolute partitioning of global methane emissions into fossil and microbial components for any given year, but we have shown that derived emissions trends are much less sensitive to these factors. The GCP suite of inversions thus finds a lower contribution from microbial sources, with roughly 1/3 of post-2006 CH_4 growth through 2020 originating from increased fossil fuel emissions (4) (*SI Appendix*, Fig. S13). Ref. 10 showed that CH_4-only atmospheric inversions are unlikely to estimate emissions consistent with atmospheric $\delta^{13}C$ data, with source partitioning largely determined by prior emission fields.

The NOAA Global Monitoring Laboratory's CarbonTracker-CH_4 (CT-CH_4) estimates global CH_4 emissions by assimilating both CH_4 and $\delta^{13}C–CH_4$ observations in a global-3D model inversion framework (10, 15). Because both models are constrained by $\delta^{13}C–CH_4$ trends, the CT-CH_4v2023 posterior largely agrees with our finding that a rise in microbial emissions has driven post-2006 CH_4 growth, though its increase is on the upper end of our estimates (Fig. 3). This is likely mostly due to CT-CH_4's inclusion of a decreasing CH_4 lifetime caused by an increasing OH sink (10, 15, 34), requiring higher emissions to reconcile the increasing atmospheric mixing ratio, as shown in our sensitivity tests. Further, our identification of a –0.0153‰ yr^{-1} trend in global mean microbial $\delta^{13}C–CH_4$ (Methods) has the effect of lowering the rise in microbial emissions by ~10 Tg yr^{-1}. This effect is not currently considered in CT-CH_4, but will be incorporated in upcoming versions (15). Ref. 14 recently estimated global emissions using $\delta^{13}C$, δD, and ^{14}C records, finding roughly equal contributions from microbial and fossil sources to CH_4 growth through 2013, though with high uncertainty in their trends. Our $\delta^{13}C$ and δD mass balances yield similar results with comparable microbial and fossil fuel emission increases through 2013 (Fig. 2). After 2013, however, fossil fuel emissions declined while microbial emission growth accelerated as the negative trend in $\delta^{13}C–CH_4$ became more pronounced (Fig. 1). A clear peak in fossil fuel emissions is harder to identify in our $\delta D–CH_4$ mass balance due to larger trend uncertainties.

Studies invoking a partial or dominant fossil fuel emission driver of recent CH_4 growth cite increased fossil fuel production as evidence (17, 43). However, production is not necessarily a reliable proxy for CH_4 emissions (44, 45) and temporal changes in this relationship remain a major source of uncertainty in bottom–up inventories (46). Accurately characterizing this link is critical for constraining fossil fuel CH_4 emissions and their trends (44). Our results suggest that despite this growth in production, fossil fuel CH_4 emissions have remained relatively stable from 1999 to 2022 (Fig. 2). We evaluated this implied improvement in emission factors by calculating how "methane intensity," defined here as CH_4 emitted per unit of natural gas CH_4 produced (5), has changed over time (Methods).

Both of our mass balance approaches suggest a 4.5% decrease in global methane intensity from 1999 to 2022 (Fig. 4). While direct observations of global methane intensity trends are lacking, satellite- and ground-based estimates from the United States and Romania over the past decade (47–49) show declines consistent with our results. Although these regional observations may not be globally representative, the consistency with our findings increases confidence that a global decline in methane intensity is realistic. Moreover, large spatial differences in methane intensity observed

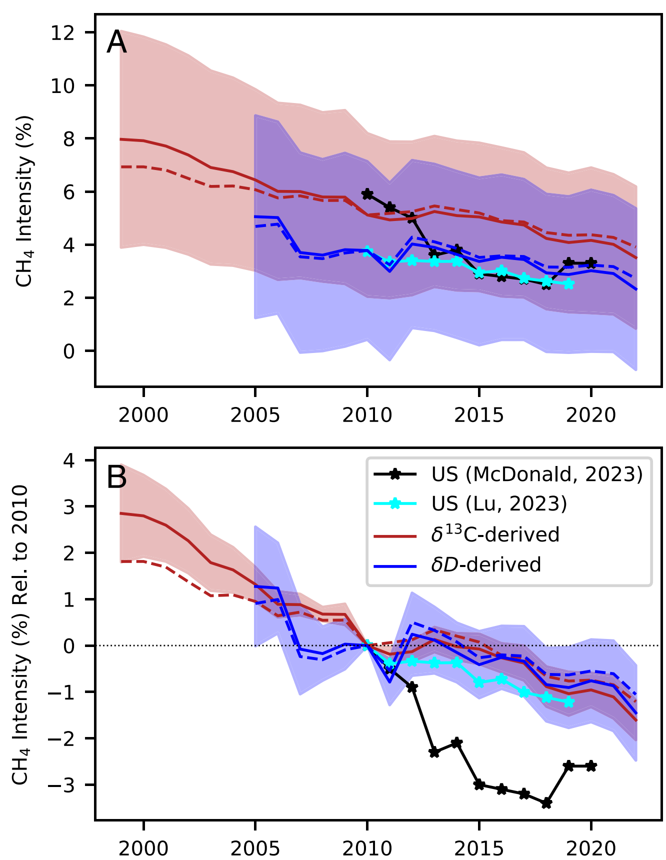

Fig. 4. Calculated methane intensity vs. existing estimates. (*A*) Total methane intensity and (*B*) change in methane intensity relative to 2010. Satellite- and ground-based estimates for the United States are shown in black (48) and cyan (47). Calculations from $\delta^{13}C–CH_4$ mass balance (red), and $\delta D–CH_4$ mass balance (blue) are shown. Two-sigma total and relative uncertainties propagated from the mass balance MC simulation are shown as the shaded areas. Dashed lines indicate methane intensity calculated assuming stable coal CH_4 emissions (S_{Coal}, Eq. **12**).

across the Middle East (45) provide evidence that large reductions are achievable, particularly through infrastructure modernization, improved management practices, and financial incentives (16). Despite uncertainty in global mean methane intensity, the consistency between our more constrained trends calculations (Fig. 4) and those derived from observational studies provides compelling evidence that fossil fuel CH_4 emissions have remained relatively stable even as production has increased.

An entirely microbial driver of post-2006 CH_4 growth likely involves significant contributions from both natural (primarily wetland) and anthropogenic (e.g., rice agriculture, landfill, ruminant) sources. While isotopic constraints cannot easily distinguish between these subcategories (3, 10, 18), bottom–up (9) and other (4) estimates of anthropogenic microbial emissions suggest a 35 Tg yr^{-1} rise between 2005–07 and 2020–22. Given that trends in anthropogenic microbial emissions are comparatively better constrained due to more consistent spatial patterns, documentation, and direct measurement (4), this implies that ~40 Tg yr^{-1} of our modeled 75 Tg yr^{-1} increase may stem from natural sources (Fig. 2). There is evidence of increasing natural microbial CH_4 fluxes in Arctic regions (50, 51), and satellite observations and inversions point to growing emissions in tropical, wetland-dominated regions (10, 15, 52–54). Furthermore, paleoclimate records show CH_4 increases of similar magnitude and pace linked to abrupt climate shifts and changes in tropical hydrology (31, 55–58), suggesting that the scale of modern wetland CH_4 increases is plausible. Wetland modeling studies specifically aimed at quantifying the impact of rising temperatures and enhanced precipitation estimate a 10 to 20 Tg yr^{-1} increase in natural wetland CH_4 emissions between 2005 to 07 and 2020 to 22 (59–61). Our results therefore suggest that either natural microbial CH_4 emission climate sensitivity is stronger than predicted, or that anthropogenic microbial emissions have increased more than bottom–up inventories suggest.

Conclusion

Our global mass balance approach and trend uncertainty analyses show that trends in global mean atmospheric δD–CH_4 and $\delta^{13}C$–CH_4 data are both highly consistent with a microbial emission driver of post-2006 CH_4 growth, with relatively stable fossil fuel emissions. While uncertainties in source signatures and sink fractionation affect the magnitude of emission estimates, they have limited influence on emission trends. We also used our δD–CH_4 record to rule out scenarios that would reconcile increased fossil fuel emissions with the observed decrease in atmospheric $\delta^{13}C$–CH_4 (i.e. reductions in pyrogenic emissions and the Cl sink). This work demonstrates the value of atmospheric δD–CH_4 and underscores the need to expand atmospheric measurements and incorporate them into 3-D models, especially at smaller spatiotemporal scales not explored in this study (e.g., refs. 23 and 62).

Our calculated decline in global methane emission intensity from the fossil fuel sector supports emerging satellite- and ground-based evidence from the United States, Europe, and the Middle East indicating that fossil CH_4 emissions are relatively stable or decreasing, underscoring the potential and need for further mitigation to achieve IPCC targets. In contrast, the concurrent rise in microbial emissions likely reflects both intensifying wetland emissions due to climate change and growing anthropogenic sources. The calculated increase in wetland emissions is consistent in both magnitude and rate with paleo reconstructions of past changes and is supported by observed shifts in tropical hydrology and Arctic warming. While further work is needed to disentangle these components, the growing contribution from climate-sensitive microbial emissions reinforces the urgency of reducing CH_4 emissions from the anthropogenic sectors we can control.

Materials and Methods

Atmospheric Measurements.

Atmospheric $\delta^{13}C$–CH_4. The NOAA Global Greenhouse Gas Reference Network (63) consists of atmospheric trace gas measurements of surface flask samples from around the world. CH_4 measurements are made weekly and are reported on the WMO X2004A mole fraction scale (64) (Fig. 1*A*). A subset of these samples is then analyzed for $\delta^{13}C$–CH_4 at INSTAAR, CU Boulder (65). Sampling sites that receive well-mixed background air from a large volume of the atmosphere are used to determine the global and zonal mean compositions of trace gas species in the reference network, known as the Marine Boundary Layer (MBL) Reference (66–68). There are 31 MBL sampling sites for CH_4 beginning in 1986 and 10 for $\delta^{13}C$–CH_4 beginning in 1998 (18). One-sigma uncertainties in global mean $\delta^{13}C$–CH_4 (<0.02‰) are determined using a 1,000-simulation MC approach that considers analytical error, site distribution, and synoptic-scale atmospheric variability (Fig. 1*B*) (67).

Atmospheric δD–CH_4. To determine global mean atmospheric δD–CH_4, we compiled measurements from four laboratories and harmonized the measurements onto a common isotopic scale using existing estimates of laboratory calibration scale offsets (*SI Appendix*, Fig. S1 and Table S1). From 2005–2009, δD–CH_4 was measured at 12 GML Global Flask Network MBL sites at submonthly resolution. The δD–CH_4 measurement system consisted of a GV (now Elementar) Isoprime isotope-ratio mass spectrometer coupled to a custom-built CH_4 trapping system, a gas chromatograph, and a pyrolysis furnace (24). Measurements of δD–CH_4 were discontinued in 2009 due to technical issues related to the analytical system and the need to prioritize resources toward other measurements at the INSTAAR Stable Isotope laboratory (SIL). Anomalous data points were flagged following an internal flagging scheme developed for trace gas isotopic measurements within the INSTAAR SIL. In total, 1118 measurements (replicates averaged when available) are included in our merged dataset (*SI Appendix*, Table S1) with estimated 1.0‰ measurement precision (3, 65).

The Max Planck Institute for Biogeochemistry (MPI-BGC) also runs an atmospheric observation network containing 15 sampling sites globally (69). Starting in 2010, MPI began measurements of atmospheric δD–CH_4 at 13 of these sites, with six considered MBL. The result is a comprehensive dataset publicly available through December 2024 with 2252 MBL measurements (69). Next, Tohoku University and the National Institute for Polar Research (TU/NIPR) make routine atmospheric measurements of δD–CH_4, with MBL data beginning in 1995 in the SH and 2005 in the NH and running through 2023 with 962 measurements available (70, 71). Finally, Utrecht University's Institute for Marine and Atmospheric Research (UU IMAU) has made atmospheric δD–CH_4 measurements at NH and SH MBL sampling sites since 1988, generating a total of 763 measurements (23, 33).

Laboratory scale comparisons. Our merged δD–CH_4 requires robust calibration scale comparisons, which are known to contain offsets up to 13‰ for δD–CH_4 (24, 33). These comparisons were established using Round Robin measurements, where a reference gas cylinder or air sample is exchanged between laboratories, and the difference between each group's measurements is used to characterize the scale offsets. We harmonized all data to the MPI scale using the offsets from ref. 24 and a recent comparison that found an offset of 11.5 ± 1.5‰ between the INSTAAR and MPI scales (*SI Appendix*, Table S1). Despite minor differences between the scale offsets reported in ref. 24 and a more recent estimate (33) (*SI Appendix*, Table S1), our emission trend results are not sensitive to the choice of scale offset applied (*SI Appendix*, Fig. S11).

Global mean δD–CH_4 and uncertainty. Due to limited δD–CH_4 data, we employed a modified version of the MC-based "Data Extension and Integration" approach (68) to calculate the global mean atmospheric δD–CH_4 and uncertainty. First, we calculated the smoothed dataset for each of the 23 MBL sampling sites (*SI Appendix*, Fig. S2*D*) using the NOAA GML curve-fitting software, which calculates weekly averages by fitting a function to the data and filtering residuals following standard parameterizations (short term filter = 150 d; long term filter = 667 d; sample interval = 7 d; number of function polynomial terms = 3; number of function harmonics = 4) (72).

The smoothed δD–CH_4 curves were binned by latitudinal blocks of equal area: −90° to −30° (Polar South, PS), −30° to 0° (Tropical South, TS), 0° to 30° (Tropical North, TN), and 30° to 90° (Polar North, PN). The average δD–CH_4 for each bin was then calculated at weekly intervals. To address occasional data gaps in latitudinal bins, we calculated a weekly mean "difference climatology" between bins, then used these differences to estimate values during intervals of missing data. Weekly intervals were chosen to capture the antiphased hemispheric seasonal cycles observed in atmospheric δD-CH_4 (*SI Appendix*, Fig. S2). For example, if data were missing for TS but available for PS and PN, the weekly mean differences (PS–TS and PN–TS) were subtracted from PS and PN, respectively, to generate two theoretical TS records, which were then averaged to create an artificial TS record. We did not use tropical bins to fill polar δD-CH_4 gaps due to sparse data in these regions. Importantly, the PN bin contained continuous data throughout our record (2005–2023), so artificial records could be generated for each bin at all timesteps (*SI Appendix*, Fig. S2). All latitudinal bins were then averaged to calculate weekly and annual global mean δD-CH_4 (Fig. 1*C*). The anomalous decrease in global mean δD-CH_4 in 2010 is likely an artifact of reduced data coverage (Fig. 1 and *SI Appendix*, Fig. S2). While this artifact influences flux estimates on interannual timescales, its effect on the long-term trend is minimal (Fig. 2).

We estimated uncertainty in global mean δD-CH_4 and its temporal trend using an MC approach (67). Uncertainty in the global mean due to biases in the distribution of MBL sites were estimated by randomly withholding two sampling sites from the global mean calculation in each MC iteration. Next, MBL sites experience variability on synoptic scales that temporarily bias site measurements from the true longitudinal mean. We estimated analytical and synoptic-scale variability by calculating the SD of the difference between measurements and the smoothed curve for each site, then randomly applying this site-specific residual SD to the data to calculate new smoothed curves in each MC iteration. Finally, we incorporated uncertainties in interlaboratory calibration scale comparisons (*SI Appendix*, Table S1). For each MC iteration, we applied random Gaussian errors to adjust the INSTAAR, UU, and TU datasets within the uncertainty of their MPI offsets. This approach generated estimates of both uncertainty in the global mean for a given year and uncertainties in the temporal trend. This uncertainty was propagated in our MC isotope mass balance analysis (Methods).

Source Signatures and Sink Fractionation. Atmospheric CH_4 emissions can be partitioned into three major source categories using isotope ratio data: Pyrogenic (wildfires and biofuel combustion), microbial (wetlands, rice paddies, ruminants, waste, etc.), and fossil (emissions related to fossil fuel production and usage and natural geologic seepage). A growing inventory of the isotopic signatures (δD-CH_4 and δ^{13}C-CH_4) of these emissions categories indicates that they are isotopically distinguishable (3, 39). We present an updated isotopic source signature database that builds on ref. 3, incorporating additional measurements and combining other recent inventories (73, 74) (*SI Appendix*, Table S2). Using these data and known spatial variability, we produced separate 1° × 1° gridded coal, oil and natural gas (ONG), pyrogenic, and microbial source signature maps for δD-CH_4 and δ^{13}C-CH_4 (*SI Appendix*, Fig. S3), which can be used in future CH_4 isotopic 3D-chemistry modeling studies. In the following subsections, we used these maps to consider important spatial and temporal variability in our flux-weighted global mean source signature calculations (Summarized in *SI Appendix*, Tables S5 and S6).

Fossil source signatures. Global mean fossil fuel source signatures were calculated by first determining the mean and SD of the ONG and coal δD-CH_4 and δ^{13}C-CH_4 for countries where data are available (*SI Appendix*, Fig. S3). Because recent ^{14}C-derived estimates of natural geologic emissions suggest that global emissions are <5 Tg yr^{-1} (31, 32), and the isotopic signature of natural geological seepage is similar to fossil fuel emissions (74), it was excluded from calculations of global mean fossil δD-CH_4 and δ^{13}C-CH_4. The annual global mean was calculated by flux-weighting these isotopic source signatures by the coal and ONG CH_4 emissions of each country reported in the EDGAR8 database (9). For countries with no source signature data, the global mean and SD for coal and ONG emissions were used. According to EDGAR8, these countries make up 46% of ONG and 24% of coal CH_4 emissions for δD-CH_4, and 34% of ONG and 14% of coal for δ^{13}C-CH_4, highlighting the importance of improving source signature measurements in undersampled regions. To test the sensitivity of our global mean calculations to spatial emission estimates, we recalculated global mean fossil source signatures using CT-CH_4 posterior emissions, which differ spatially from EDGAR8 fluxes (*SI Appendix*, section 2) (15). This shifted the absolute values but preserved their trends (*SI Appendix*, Fig. S5), resulting in small changes to the absolute magnitude of emissions but no impact on emission trends uncertainties (*SI Appendix*, Fig. S11).

Flux-weighting by annual emissions provides estimates of temporal variability in global mean fossil fuel source signatures. These considerations are crucial as fossil fuel production and associated emissions have varied over time among countries with differing isotopic source signatures. Our global mean trend estimates also consider how the rise in shale gas production, which contains CH_4 enriched in both ^{13}C and D relative to other fossil sources (39, 74), has caused the mean ONG source signatures of the United States, China, Argentina, and Canada to increase over time (*SI Appendix*, Fig. S4 and section 1).

Uncertainties in global mean fossil fuel source signatures were estimated using an MC approach, where the δD-CH_4 and δ^{13}C-CH_4 of each country's ONG and coal source signatures were randomly varied using a Gaussian distribution defined by the SE of the mean source signatures for each country. This approach quantifies both the absolute global fossil fuel source signature and uncertainty over time. We find that global mean fossil δ^{13}C-CH_4 increased from −44.3 to −43.7‰ from 1999 to 2022 (*SI Appendix*, Fig. S8*A*). Although global mean fossil δ^{13}C-CH_4 is somewhat uncertain (±0.2‰), the 0.60‰ increase across the analysis period is less so (±0.14‰) (*SI Appendix*, Fig. S8*A*). Global fossil δD-CH_4 changed little across the analysis period, remaining at roughly −194.9 ± 0.6‰, with a minor net increase of 1.3 ± 0.7‰ (*SI Appendix*, Fig. S8*B*).

Microbial source signatures.

δD–CH_4. There is a linear relationship between the δD-CH_4 of microbial CH_4 emissions and the annual mean δD-H_2O of local environmental water (25, 75). Although the regression is based on wetland and inland water samples, other microbial CH_4 sources likely show similar fractionation, as methanogenesis universally draws hydrogen from local water (75). However, CH_4 emitted from landfills undergoes partial consumption and isotopic fractionation prior to atmospheric release (76). Using available landfill δD-CH_4 measurements, we derived a landfill-specific regression (δD-$CH_{4,\,Landfill}$ = (0.757 ± 0.436) * δD-H_2O – 245.8 ± 23.7) (*SI Appendix*, Fig. S6). The slope of this updated regression is comparable to that reported by ref. 25, suggesting mechanistically similar incorporation of the local environmental δD-H_2O signal, but the intercept is higher due to the preferential consumption of $^{12}CH_4$ by methanotrophs prior to atmospheric release (76) (*SI Appendix*, Fig. S6).

The wetland regression from ref. 75 (δD-$CH_{4,\,Wetland}$ = (0.6088 ± 0.072) * δD-H_2O – 285.7 ± 6.9) and our landfill CH_4-specific regression were paired with high-resolution spatial maps of the annual mean δD-H_2O of precipitation (77) to create predictive 1° × 1° maps of the δD-CH_4 of landfill and all other microbial CH_4 emissions (*SI Appendix*, Fig. S3*D*). Next, we determined the global annual flux-weighted mean δD-CH_4 of each microbial subcategory (*SI Appendix*, Table S4) and total microbial emissions using these spatial maps and CT-CH_4 posterior 1° × 1° gridded emission fields (15). We found a global mean microbial source signature of −304.7‰ (*SI Appendix*, Fig. S8*D*) and a statistically significant +0.11 ‰/yr linear temporal trend (R^2 = 0.541) (*SI Appendix*, Fig. S7) that was driven by enhanced microbial emissions from deuterium-enriched tropical regions.

We estimated uncertainty in the global mean using an MC approach where for each iteration, updated source signature maps were created by varying the δD-CH_4 vs. δD-H_2O regressions within their error bounds (*SI Appendix*, Fig. S6) (75) and subcategory proportions were varied by ±10% (*SI Appendix*, Fig. S8*D*). This resulted in a global mean SD of 8.2‰. To account for uncertainties in trends in the fractional contributions and source signatures of each microbial subcategory, we varied the calculated global mean microbial δD-CH_4 trend by ±25% in each MC input.

δ^{13}C–CH_4. The global mean δ^{13}C-CH_4 of microbial CH_4 emissions is impacted by the varying distribution of isotopically distinguishable C3 and C4 vegetation (78). The δ^{13}C-CH_4 of microbial emissions is also expected to be decreasing over time due to the Suess Effect, in which ^{13}C-depleted fossil CO_2 emissions are incorporated into the organic carbon that is ultimately metabolized to CH_4 by methanogens (20, 79).

To calculate global mean microbial δ^{13}C-CH_4, we flux-weighted the global mean δ^{13}C-CH_4 of each emission subcategory (*SI Appendix*, Table S4). We used the annual global mean and uncertainty of wetland δ^{13}C-CH_4 from ref. 20, which considers both the Suess Effect and spatial variability in source signature (20). For ruminants, we used the global mean and uncertainty calculated by ref. 40, which considers both the Suess Effect and the changing C3/C4 composition of ruminant diets. Due to limited measurements of CH_4 emitted from rice agriculture, wild animals, waste, and termites, we used the mean and SD of available measurements

(*SI Appendix*, Fig. S8*D*) and imposed a decreasing trend of 0.024‰ yr^{-1} on each to consider the Suess Effect (20) (*SI Appendix*, Fig. S7). Flux-weighting the subcategories using CT-CH_4 posterior emissions resulted in a global mean value of −62.0‰ with a trend of −0.0153‰ yr^{-1} (*SI Appendix*, Fig. S7).

Uncertainty in global mean $\delta^{13}C$-CH_4 was estimated using an MC approach where the contribution of different microbial subcategories to total microbial emissions was varied by ±10% and global mean subcategory isotopic signatures were varied within their uncertainties (*SI Appendix*, Fig. S8*D*), resulting in a global mean SD of 1.3‰. We also considered uncertainty in the calculated global mean microbial $\delta^{13}C$-CH_4 trend by varying it ± 25% in each MC iteration.

Pyrogenic source signatures.

δ^{13}*C–CH*$_4$. The $\delta^{13}C$-CH_4 of pyrogenic emissions is also sensitive to whether combusted biomass was derived from C3 or C4 photosynthetic pathways (5). We combined our calculated mean and SD of C3- and C4-derived pyrogenic $\delta^{13}C$-CH_4 and the global C4 map from ref. 80 to create a gridded 1° × 1° map of pyrogenic $\delta^{13}C$-CH_4 (*SI Appendix*, Fig. S3*G*). To determine the global mean, we flux-weighted this map using gridded annual emission fields from CT–CH_4 (15) (*SI Appendix*, Fig. S8*E*). Ordinary least squares (OLS) regression analyses reveal no statistically significant temporal trend in the calculated global mean ($P = 0.17$), suggesting a negligible influence of potential expansion of C3 vegetation (80). We therefore use the CT–CH_4-weighted global mean of all years (−23.1‰) for each year of our mass balance calculations. We estimated uncertainty in global mean pyrogenic $\delta^{13}C$-CH_4 using an MC approach where the $\delta^{13}C$-CH_4 of C3- and C4-derived pyrogenic emissions were randomly varied within the calculated uncertainties. The resulting SD of all MC iterations and all years was 2.5‰.

δ*D–CH*$_4$. To determine global mean pyrogenic δD-CH_4, we considered the established regression between pyrogenic δD-CH_4 and the annual mean δD-H_2O of local water from ref. 81 ($\delta D\text{-}CH_{4,\,pyrogenic} = (1.16 \pm 0.18) * \delta D\text{-}H_2O - 177 \pm 13$). This regression and the gridded map of annual mean precipitation δD-H_2O (77) were used to calculate a predictive 1°x1° spatial map of pyrogenic δD–CH_4 (*SI Appendix*, Fig. S3*C*). We then flux-weighted the source signature map with CT-CH_4 posterior pyrogenic emission fields to determine the global mean for each year (*SI Appendix*, Fig. S8*F*). With the exception of 2021, a year with anomalously high fire activity in Siberia (82), OLS regression analyses reveal no temporal trend ($P = 0.192$), so we use the mean of all years and iterations in our mass balance calculations (−228‰). Uncertainty in the global mean pyrogenic δD-CH_4 was determined using an MC approach where the pyrogenic δD-CH_4 map was recalculated for each iteration using a new regression based on the regression uncertainty (81). The global mean one-sigma uncertainty is defined as the SD of the flux-weighted mean of all iterations and years (7.6‰).

Sink Fractionation. Atmospheric $\delta^{13}C$-CH_4 and δD-CH_4 are modified by fractionation during atmospheric CH_4 destruction (*SI Appendix*, Table S3). The largest CH_4 sink is reaction with OH, and smaller contributions from soil uptake, reaction with Cl, and stratospheric loss (4). Although some values are uncertain, these uncertainties do not affect our mass balance trend analyses (Methods).

Global Mass Balance.

One-box model framework. We estimated global microbial and fossil fuel CH_4 emissions with an atmospheric one-box model (i.e. global mass balance) using our global annual mean stable isotope and CH_4 mole fraction datasets (Fig. 1) (3, 5, 58, 83). We calculated the $\delta^{13}C$-CH_4 and δD-CH_4 of the global CH_4 source to derive two independent estimates of the CH_4 budget. We first calculated the global atmospheric CH_4 burden, including $^{13}CH_4$ burden (X^{13C}), CH_3D burden (X^D), and all CH_4 isotopologues (X_n):

$$X_n = X_{CH4} * \frac{m_A}{M_A} * M_{CH4}, \quad [1]$$

$$X^{13C} = \left(\delta^{13}C + 1\right) * R_{13Cstd} * X_n, \quad [2]$$

$$X^D = (\delta D + 1) * R_{Dstd} * X_n, \quad [3]$$

where m_A is the mass of the atmosphere (5.15 × 10^{21} g), M_A and M_{CH4} are the molar masses of dry air (28.97 g mol^{-1}) and CH_4 (16.04 g mol^{-1}), and X_{CH4} is the global mean annual atmospheric CH_4 mole fraction (28). In Eqs. **2** and **3**, R_{13Cstd} is the VPDB standard for $\frac{^{13}C}{^{12}C}$ and R_{Dstd} is the VSMOW standard for $\frac{D}{H}$. $\delta^{13}C$ and δD are annual global mean atmospheric $\delta^{13}C$-CH_4 and δD-CH_4, respectively. We then calculated the global CH_4 source strength for all isotopologues (index n) using the g mass balance equation:

$$\frac{dX_n}{dt} = S - \frac{X_n}{L}, \quad [4]$$

where *S* is the annual global CH_4 source strength (g yr^{-1}), and L is the lifetime (9 y) (84). Next, we calculated the global $^{13}CH_4$ and CH_3D source strength in g yr^{-1} (S_{13C} and S_D, respectively):

$$\frac{dX_n^{13C}}{dt} = S_{13C} - \frac{X_n^{13C}}{L} * \alpha_{13C}, \quad [5]$$

$$\frac{dX_n^{D}}{dt} = S_{D} - \frac{X_n^{D}}{L} * \alpha_{D}, \quad [6]$$

where α_{13C} and α_D are the inverse of the total sink kinetic isotope effect for $\delta^{13}C$-CH_4 and δD–CH_4 (1/KIE, *SI Appendix*, Table S3). The output of Eqs. **4**–**6** were used to calculate the global isotopic source signatures, $\delta^{13}C_S$ and δD_S, for each year of the mass balance calculations.

$$\delta^{13}C_S = \left(\frac{R_S^{13C}}{R_{13Cstd}} - 1\right), \quad [7]$$

$$\delta D_S = \left(\frac{R_S^{D}}{R_{Dstd}} - 1\right), \quad [8]$$

where R_S^{13C} is $\frac{S_{13C}}{S}$, and R_S^D is $\frac{S_D}{S}$. We then used the following three isotope mass balance equations to calculate the microbial and fossil components of the global CH_4 source using the empirically derived global CH_4 source (S) and its stable isotopic signatures ($\delta^{13}C_S$ and δD_S).

$$S = S_{Mic} + S_{Pyro} + S_{FF} + S_{Geo}, \quad [9]$$

$$S * \delta^{13}C_S = S_{Mic} * \delta^{13}C_{Mic} + S_{Pyro} * \delta^{13}C_{Pyro} + S_{FF} * \delta^{13}C_{FF} + S_{Geo} * \delta^{13}C_{Geo}, \quad [10]$$

$$S * \delta D_S = S_{Mic} * \delta D_{Mic} + S_{Pyro} * \delta D_{Pyro} + S_{FF} * \delta D_{FF} + S_{Geo} * \delta D_{Geo}, \quad [11]$$

where the subscripts Mic, Pyro, FF, and Geo represent the source signatures and source strengths of microbial, pyrogenic, fossil fuel, and natural geologic emissions, respectively. We assumed that the $\delta^{13}C$-CH_4 and δD-CH_4 of natural geologic emissions equal our estimates of global mean fossil fuel source signatures, and that S_{Geo} was stable at 5 Tg yr^{-1} (31, 32). Using our calculated global mean source signatures (*SI Appendix*, Fig. S8) and annual pyrogenic emissions from GFEDv4.1 for S_{Pyro}, we solved Eqs. **9**–**11** to generate two independent estimates of global microbial and fossil fuel emissions (Fig. 2).

We estimated uncertainty in the global CH_4 budget using a 1,000-simulation MC approach, which allows for the key separation of flux magnitude and trend uncertainties. The approach includes uncertainties in the global mean and trends in source signatures (*SI Appendix*, Figs. S4–S8) and atmospheric $\delta^{13}C$-CH_4 and δD-CH_4 (Fig. 1 *B* and *C*). In each MC iteration, annual-scale time series are generated for each variable by randomly varying mean values and temporal trends based on the total and trend uncertainties described in Table 1. This results in 1,000 microbial and fossil fuel emission time series with varying absolute values and trends (*SI Appendix*, Fig. S9) from which total and trend uncertainty are calculated separately (Fig. 2).

Methane intensity. We calculated methane intensity, or fugitive emission rate (5), following Eq. **12**:

$$\text{Methane Intensity} = \frac{S_{FF} - S_{Geo} - S_{Oil} - S_{Coal}}{NG_P}, \quad [12]$$

where S_{Oil} represents global oil CH_4 emissions and was determined using established conversion factors between oil production and associated CH_4 emissions (5). Although we assume stable S_{Oil} emission factors, any improvements would have minimal impact on our methane intensity calculations because oil CH_4 emissions make up only ~10% of total fossil emissions (5). Global coal CH_4 emissions (S_{Coal}) were obtained from the EDGAR8 inventory. Although coal emissions are generally more certain in emission inventories (43), we also calculated methane intensity assuming stable coal emissions, rather than the doubling

(23.8 Tg yr^{-1} increase) from 1999 to 2022 indicated by the EDGAR8 inventory (9), and found minimal impact on the calculated negative trend (Fig. 4). Natural gas CH_4 production (NG_P) was determined by multiplying global natural gas production data (*eia.gov*) by global mean natural gas CH_4 composition by weight (86%) (5). Calculations were made for each year of our global mass balance (Fig. 4). Absolute and relative methane intensity uncertainties were estimated by using the fossil fuel emission output matrix from our MC uncertainty analysis (*SI Appendix*, Fig. S9).

Sensitivity Tests. We tested the sensitivity of our analyses to several underconstrained elements related to atmospheric $\delta^{13}C$–CH_4 and δD–CH_4 interpretation. First, we evaluated the impact of calculated temporal trends in microbial and fossil fuel source signatures (Methods) by solving Eq. **9** through **11** using stable source signatures (mean of all years; Table 1 and *SI Appendix*, Fig. S11). Next, we considered the sensitivity of our analyses to changes in the Cl sink and pyrogenic emissions. The global sink fractionation for each isotope was calculated using Eq. **13**:

$$KIE_{net} = (S_{OH}) * (KIE_{OH}) + (S_{strat}) * (KIE_{strat}) + (S_{soil}) * (KIE_{soil}) + (S_{cl}) * (KIE_{cl}), \quad [\mathbf{13}]$$

where S is the fraction that each sink contributes to the total sink (*SI Appendix*, Table S3). To test the impact of a reducing Cl sink, we imposed a linear reduction in S_{cl} from 0.035 to 0.011 over 2005-2022 and adjusted the other three sink fractions proportionally to keep lifetime stable. We then solved our mass balance with this updated sink and compared the resulting emissions with our baseline scenario where sink fractionation factors remain stable (Fig. 3 and *SI Appendix*, Fig. S10). The Cl sink reduction to 0.011 was determined to be the minimum reduction required to shift the $\delta^{13}C$ mass balance results to a fossil fuel-leaning driver of recent CH_4 growth. Similarly, we tested the scenario where the OH sink, and thus total sink, increases in line with recent estimates (34, 35) by imposing a linear increase of 0.3% yr^{-1} beginning in 2005. Eq. **13** was used to calculate the alternate sink fractionation factors, and the lifetime was adjusted proportionally to reflect the increase in the total sink. We tested a third scenario where pyrogenic emissions linearly decreased from 29 Tg yr^{-1} by 30% (21, 38) and separately to 2 Tg yr^{-1} from 2005 to 2022, solving Eq. **9** through **11** for fossil fuel and microbial emissions (Fig. 3 and *SI Appendix*, Fig. S10).

Data, Materials, and Software Availability. Atmospheric CH_4 mole fraction data are available from the NOAA Global Monitoring Laboratory (GML) at https://gml.noaa.gov/data/data.php (63). The updated version of the methane isotope source signature database used in this study is archived at NOAA GML under DOI: 10.15138/92gg-ey58 (85). Global mean $\delta^{13}C$–CH_4 and δD–CH_4 values and their associated uncertainties used in the box model analyses, as well as all site-level δD–CH_4 measurements used to calculate the global mean, are available from NOAA GML at 10.15138/setb-jy31 (86). Code related to global mean annual source signature and atmospheric δD–CH_4 calculations, Monte Carlo analyses, sensitivity tests, figure generation, and methane intensity calculations, along with the input data required to run these scripts, is publicly available on Zenodo at 10.5281/zenodo.17238656 (87).

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